

# AstroLight: A Late-Fusion Deep Learning Architecture for Multi-Class Exoplanet Transit Signal Classification

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## ABSTRACT

*Automated classification of Kepler Objects of Interest (KOIs) is a fundamental challenge in the search for exoplanets. While existing deep learning pipelines (AstroNet, ExoMiner) achieve high accuracy on binary planet-candidate vs. false-positive tasks, multi-class false-positive discrimination remains underexplored. We present AstroLight, a multi-task late-fusion architecture that simultaneously classifies KOIs into five physically-motivated categories: genuine transits, stellar eclipses, centroid offsets, background noise, and ephemeris matches. Our system fuses global and local phase-folded light curve representations with 12-dimensional stellar catalog features via three parallel 1D-ResNet branches and a residual late-fusion head. Training with Focal Loss ( $\gamma=2.0$ ) achieves 79.93% five-class accuracy and a weighted F1 of 0.8041, with a transit F1 of 0.9254, substantially outperforming a CrossEntropy baseline. We provide a detailed failure analysis of the Centroid Offset over-prediction bias and the Ephemeris Match data starvation problem, and discuss the fundamental limits imposed by extreme class imbalance.*

**Index Terms** — Exoplanet detection, Kepler mission, multi-class classification, convolutional neural networks, focal loss, light curves, false-positive discrimination.

## I. INTRODUCTION

The NASA Kepler Space Telescope (2009–2018) photometrically monitored over 150,000 stars, producing a rich archive of light curves from which thousands of exoplanet candidates were identified via the transit method. However, the identification of genuine planet transits among a large population of astrophysical false positives — eclipsing binaries, background contamination, centroid offsets, and instrument artefacts — remains a significant bottleneck in the planet-candidate vetting pipeline.

Early automated methods such as the Box Least Squares (BLS) periodogram flagged periodic flux dimmings but could not differentiate their physical origin. Machine learning approaches subsequently demonstrated that convolutional neural networks (CNNs) could match or exceed expert-level accuracy on the binary classification task of planet candidate (PC) vs. false positive (FP). Shallue & Vanderburg [1] introduced AstroNet, a 1D-CNN operating on phase-folded light curves that achieved 97.5% precision on the Kepler DR24 catalog. Valizadegan et al. [2] later proposed ExoMiner, a more expressive architecture incorporating additional diagnostic inputs, reaching an AUC of 0.984.

Despite these advances, binary classification conflates physically distinct false-positive classes into a single label, losing information critical for follow-up prioritisation and stellar population studies. A centroid offset — indicating the transit originates from a background source — requires a different follow-up strategy than an ephemeris match (period coincidence with a known planet), yet both are labelled 'false positive' in binary

systems.

This work presents **AstroLight**, a five-class classifier developed through an iterative research process that began as a hackathon prototype and evolved into a multi-task late-fusion deep learning system. Our primary contributions are: (1) a three-branch late-fusion architecture that jointly processes global light curves, local transit windows, and stellar catalog parameters; (2) a multi-task learning objective combining classification, transit parameter regression, and confidence estimation; (3) a systematic investigation of loss function design for extreme class imbalance; and (4) a rigorous failure analysis revealing fundamental limitations of current approaches for rare signal classes.

## II. RELATED WORK

### A. Classical Vetting Methods

Traditional Kepler vetting relied on the Robovetter pipeline [3], a rule-based decision tree operating on diagnostic metrics from the Threshold Crossing Event (TCE) catalog. While interpretable, Robovetter's fixed thresholds were brittle across stellar populations and required manual tuning per data release.

### B. Binary Deep Learning Classifiers

AstroNet [1] demonstrated that a nine-layer 1D-CNN applied to globally and locally phase-folded light curves could achieve near-expert accuracy. Its global view (2001 bins over  $[-0.5, 0.5]$  orbital phase) captures the overall light curve morphology, while the local view (201 bins centred on the transit) captures fine transit structure. This dual-view approach has since become standard.

Ansdell et al. [4] extended this to Kepler DR25 and demonstrated transfer learning across different stellar populations. Yu et al. [5] incorporated additional stellar parameters, foreshadowing the multi-modal fusion approach we develop here. ExoMiner [2] achieved state-of-the-art binary classification by combining multiple diagnostic time series with a hierarchical feature extraction scheme, though it remains a binary classifier.

### C. Multi-Class Approaches

Multi-class vetting has received limited attention. Dattilo et al. [6] explored three-class classification (PC, EB, BEB) on Kepler data, achieving moderate performance. To our knowledge, five-class discrimination incorporating ephemeris matching as a separate category has not been previously reported in the peer-reviewed literature, making direct quantitative comparison with prior work imperfect due to task mismatch.

Table I: Comparison of selected published transit classification methods. \* indicates binary classification; AstroLight performs 5-class discrimination.

| Method                                    | Classes | Accuracy | AUC / F1  | Features Used         |
|-------------------------------------------|---------|----------|-----------|-----------------------|
| <a href="#">AstroNet [1] (2018)</a>       | 2*      | 97.5%    | AUC 0.960 | Global + Local views  |
| <a href="#">ExoMiner [2] (2022)</a>       | 2*      | ~98%     | AUC 0.984 | Multi-diag. + stellar |
| <a href="#">Ansdell et al. [4] (2018)</a> | 2*      | 97.8%    | AUC 0.950 | Global + Local views  |
| <a href="#">Dattilo et al. [6] (2019)</a> | 3*      | 88.4%    | F1 0.843  | Global + Local views  |

|                                   |   |       |            |                      |
|-----------------------------------|---|-------|------------|----------------------|
| <b>AstroLight CE (ours)</b>       | 5 | 72.1% | W-F1 0.752 | Global+Local+Stellar |
| <b>AstroLight FL (ours, best)</b> | 5 | 79.9% | W-F1 0.804 | Global+Local+Stellar |

### III. DATASET AND PREPROCESSING

#### A. The NASA Kepler KOI Catalog

Our dataset is derived from the NASA Kepler Objects of Interest (KOI) cumulative catalog (koi\_cumulative), augmented with manually-curated signal-class labels. Each entry corresponds to a unique Kepler Input Catalog (KIC) identifier linked to a downloaded light curve (.npz) file containing barycentric-corrected timestamps and normalised flux values.

The final labeled dataset contains **9,564 entries** spanning five classes. Table II details the class distribution, which exhibits severe imbalance: genuine transits constitute nearly half the dataset, while Ephemeris Match accounts for only 1.3%.

Table II: Dataset class distribution across all splits.

| Class           | Label | Total Count  | Train (70%) | Val (15%) | Test (15%) | % of Total |
|-----------------|-------|--------------|-------------|-----------|------------|------------|
| Transit         | 0     | <b>4,720</b> | 3,307       | 708       | 709        | 49.4%      |
| Stellar Eclipse | 1     | <b>2,209</b> | 1,547       | 331       | 331        | 23.1%      |
| Not Transit     | 2     | <b>1,300</b> | 909         | 196       | 195        | 13.6%      |
| Centroid Offset | 3     | <b>1,207</b> | 844         | 182       | 181        | 12.6%      |
| Ephemeris Match | 4     | <b>128</b>   | 87          | 22        | 19         | 1.3%       |
| <b>Total</b>    | —     | <b>9,564</b> | 6,694       | 1,439     | 1,435      | 100%       |

#### B. Preprocessing Pipeline

Each raw light curve undergoes the following standardised pipeline before being presented to the model:

- **Outlier removal:** 5-sigma clipping on the flux time series.
- **Phase folding:** timestamps are mapped to orbital phase  $\phi = ((t - t_0) / P) \bmod 1$ , then shifted to  $[-0.5, 0.5]$  via  $\phi - 1$  for  $\phi > 0.5$ .
- **Global view:** median-binned representation over the full phase range  $[-0.5, 0.5]$  into 1,001 uniform bins. Captures full light curve morphology.
- **Local view:** median-binned representation over  $[-W, W]$  where  $W = 2.5 \times (\text{duration}/\text{period})$ , also into 1,001 bins. Captures transit ingress/egress structure.
- **Normalisation:** each view is median-subtracted and std-normalised independently.
- **Stellar features:** 12 catalog features (koi\_period, koi\_duration, koi\_depth, koi\_prad, koi\_teq, koi\_steff, koi\_slogg, koi\_smet, koi\_bin\_oedp\_sig, koi\_fwm\_stat\_sig, koi\_dicco\_msky, koi\_model\_snr) are Z-score normalised using training-set statistics. Missing values are imputed with the per-column training median.

Training samples are augmented online with three stochastic operations each applied with probability 0.5: additive Gaussian noise ( $\sigma=0.008$ ), circular phase shift (up to 30 bins), and multiplicative flux scaling ( $[0.98, 1.02]$ ). Augmentation is applied after caching the base preprocessed views, ensuring the heavy preprocessing computation is performed only once per sample.

## IV. ARCHITECTURE

### A. Project Evolution

AstroLight evolved through two distinct phases. The initial prototype (Model 1) was developed under hackathon time constraints as a binary classifier using a Bidirectional LSTM (BiLSTM) with an additive attention mechanism applied to the raw light curve sequence. While functional, the BiLSTM could not capture the multi-scale structure of transit signals and lacked the stellar context necessary for fine-grained false-positive discrimination.

Recognising these limitations, the problem statement was revised from binary to five-class multi-label discrimination, and Model 2 was designed ground-up as a late-fusion multi-task architecture.

### B. Model 1: BiLSTM + Attention (Baseline)

The baseline model processes the global-view light curve as a sequence of flux values through a two-layer Bidirectional LSTM (hidden size 128, dropout 0.3). A learned additive attention mechanism aggregates hidden states into a fixed-length context vector, which is then passed through a 2-layer MLP to produce class logits. This model was trained on binary (transit / not-transit) labels only and served as a proof-of-concept.

### C. Model 2: Late-Fusion Multi-Task Architecture

The production model, **ExoplanetLateFusionModel**, processes three independent input streams in parallel before fusing their representations:

- **Global Branch:** A 1D residual CNN (GlobalViewBranch) applied to the 1,001-bin global view. The branch produces a sequence of 32 temporal feature vectors of dimension 256, which are globally average-pooled to a 256-d embedding.
- **Local Branch:** An identical 1D ResNet (LocalViewBranch) applied to the 1,001-bin local view centred on the transit. Output is similarly pooled to a 256-d embedding.
- **Stellar Branch:** A 3-layer fully-connected network (StellarFeaturesBranch) mapping the 12-d catalog feature vector to a 128-d embedding via ReLU activations and BatchNorm.

The three embeddings are concatenated into a 640-d joint representation and passed through a **ResidualLateFusionHead**: a 2-layer MLP ( $640 \rightarrow 512 \rightarrow 256$ ) with a parallel linear shortcut ( $640 \rightarrow 256$ ), BatchNorm, ReLU, and Dropout ( $p=0.4$ ). The residual connection stabilises gradient flow through the fusion stage.

The 256-d refined representation branches into three multi-task heads: (1) a **ClassifierHead** ( $256 \rightarrow 64 \rightarrow 5$ ) producing class logits; (2) a **regression head** ( $256 \rightarrow 4$ ) predicting log-scaled transit depth, duration, period, and epoch; and (3) a **confidence head** ( $256 \rightarrow 1$ , sigmoid) producing a scalar transit probability. Temperature scaling is applied to class logits via a learnable parameter.

Table III: Model 2 architecture summary.

| Component        | Input Shape | Output Shape | Key Details          |
|------------------|-------------|--------------|----------------------|
| GlobalViewBranch | (B,1,1001)  | (B,32,256)   | 1D ResNet, Self-Attn |

|                       |              |            |                        |
|-----------------------|--------------|------------|------------------------|
| LocalViewBranch       | (B,1,1001)   | (B,32,256) | 1D ResNet, Self-Attn   |
| StellarFeaturesBranch | (B,12)       | (B,128)    | FC 12→64→128, BN       |
| GAP + Concat          | 3 embeddings | (B,640)    | Global avg pool + cat  |
| ResidualFusionHead    | (B,640)      | (B,256)    | MLP+shortcut, drop 0.4 |
| ClassifierHead        | (B,256)      | (B,5)      | FC 256→64→5            |
| RegressionHead        | (B,256)      | (B,4)      | FC 256→4, masked MSE   |
| ConfidenceHead        | (B,256)      | (B,1)      | FC 256→1, sigmoid      |

## V. TRAINING STRATEGY

### A. Optimiser and Scheduler

All models are trained with the **AdamW** optimiser ( $\text{lr}=1\text{e-}3$ ,  $\text{weight\_decay}=1\text{e-}4$ ) and a **OneCycleLR** cosine annealing scheduler ( $\text{max\_lr}=3\text{e-}3$ ,  $\text{pct\_start}=0.15$ ). Gradient norms are clipped to 5.0. Early stopping monitors validation ROC AUC with patience=20 epochs over a maximum of 80 epochs.

### B. Multi-Task Loss

The joint training objective is:

$$L = L_{\text{class}} + 0.4 * L_{\text{reg}} + 1.0 * L_{\text{conf}}$$

- **L\_class**: Classification loss (CrossEntropy or Focal Loss, see §V.C).
- **L\_reg**: Masked MSE on {depth, duration, period, epoch} — computed only on ground-truth transit samples (class 0) to avoid contaminating regression with false-positive geometry.
- **L\_conf**: Binary Cross Entropy on the confidence head (target=1 for transit, 0 otherwise).

### C. Handling Class Imbalance

The dataset's extreme imbalance (Transit:Ephemeris = 38:1) required a carefully designed correction strategy. Three mechanisms were evaluated:

- **WeightedRandomSampler**: Training batches are drawn with replacement using per-sample weights =  $1/\text{class\_count}$ , with an  $\text{ephemeris\_boost}=2.0\times$  multiplier for Ephemeris Match samples. This operates at the data level.
- **Inverse-frequency class weights in CrossEntropy** (Run 1 baseline): Full inverse-frequency weights are passed into CrossEntropyLoss. This operates at the loss level.
- **Focal Loss** [7] (Run 2, best model): Replaces CrossEntropy with  $\text{FL}(p_t) = -(1-p_t)^\gamma * \log(p_t)$ ,  $\gamma=2.0$ . No class\_weights are passed — the Sampler handles data-level imbalance while Focal Loss handles hard-example focus.

**Critical finding**: Stacking all three corrections simultaneously (Sampler + class\_weights + Focal Loss + ephemeris\_boost) caused catastrophic model collapse, with the model defaulting to predicting Ephemeris Match for almost all inputs (accuracy 24.4%). The over-correction created an effective 40x boost for the rarest class, destabilising the optimisation landscape entirely. The resolution was to use Sampler + Focal Loss only (no class\_weights), treating them as operating on orthogonal axes: data distribution vs. hard-example weighting.

## VI. EXPERIMENTS AND RESULTS

### A. Experimental Setup

All experiments use a stratified 70/15/15 train/val/test split with random\_state=42 applied to the full labeled catalog. Validation-set ROC AUC (one-vs-rest, macro-averaged) is used for model selection. Test-set metrics are reported on the best checkpoint only. All experiments run on a single NVIDIA GPU (CUDA). Preprocessing caching reduces per-epoch data loading time from ~166s (epoch 1, cold) to ~14s (subsequent epochs, warm cache).

### B. Run 1: CrossEntropy with Inverse-Frequency Weights

The baseline configuration uses standard CrossEntropyLoss with full inverse-frequency class weights and WeightedRandomSampler. Training converged at epoch 34 with a best validation ROC AUC of 0.8992. While Transit F1 was acceptable (0.867), the Centroid Offset class exhibited severely degraded precision (0.364), indicating the model was using it as a 'default rejection bucket' for ambiguous signals. Ephemeric Match F1 was near zero (0.050).

### C. Run 2: Focal Loss (Best Model)

Replacing CrossEntropy with Focal Loss (gamma=2.0, no class\_weights) produced significant gains across all major classes. Transit F1 improved from 0.867 to 0.925 (recall 93.7%), and Centroid Offset F1 jumped from 0.475 to 0.624 — confirming that the Focal Loss's hard-example downweighting directly reduced the over-prediction bias. Accuracy improved from 72.1% to 79.9%.

Table IV: Per-class and aggregate performance metrics on the held-out test set (1,435 samples). Best values per metric are bolded.

| Class / Metric  | Precision (CE) | Recall (CE) | F1 (CE)       | Precision (FL) | Recall (FL) | F1 (FL)       | Support |
|-----------------|----------------|-------------|---------------|----------------|-------------|---------------|---------|
| Transit         | 0.9703         | 0.7842      | 0.8674        | 0.9146         | 0.9365      | <b>0.9254</b> | 709     |
| Stellar Eclipse | 0.8538         | 0.6526      | 0.7397        | 0.8429         | 0.6647      | <b>0.7432</b> | 331     |
| Not Transit     | 0.6618         | 0.7026      | 0.6816        | 0.7181         | 0.6923      | <b>0.7050</b> | 195     |
| Centroid Offset | 0.3636         | 0.6851      | 0.4751        | 0.5650         | 0.6961      | <b>0.6238</b> | 181     |
| Ephemeric Match | 0.0328         | 0.1053      | <b>0.0500</b> | 0.0541         | 0.1053      | <b>0.0714</b> | 19      |
|                 |                |             |               |                |             |               |         |
| Accuracy        |                |             | 0.7213        |                |             | <b>0.7993</b> | 1435    |
| Macro avg F1    |                |             | 0.5628        |                |             | <b>0.6138</b> | 1435    |
| Weighted F1     |                |             | 0.7524        |                |             | <b>0.8041</b> | 1435    |
| Val ROC AUC     |                |             | 0.8992        |                |             | <b>~0.90+</b> | —       |

## VII. FAILURE ANALYSIS

### A. The Centroid Offset Gravity Well

In Run 1, the Centroid Offset class exhibited a pathological behaviour we term the 'gravity well effect': the model predicted centroid\_offset for a disproportionate number of non-centroid samples, yielding precision of only 0.364 despite reasonable recall (0.685). This manifested as Transit recall of only 78.4% — approximately one in five real transits was incorrectly classified as centroid\_offset.

Analysis of the confusion matrix revealed that the model had learned to treat centroid\_offset as a catch-all rejection class. This is a known failure mode of weighted cross-entropy training: when class weights are large and the minority-class decision boundary is uncertain, the optimiser finds a local minimum where the high-weight class absorbs uncertain predictions.

Focal Loss directly addresses this by applying a  $(1-p_t)^2$  modulating factor that reduces the gradient contribution of over-confident incorrect predictions. After switching to Focal Loss, Centroid Offset precision improved from 0.364 to 0.565 (a 55% relative improvement), and Transit recall recovered to 93.7%.

## B. Ephemeris Match: The Data Starvation Problem

Ephemeris Match — where a KOI's orbital period coincidentally matches a known planet in the catalog — is the most challenging class for two fundamental reasons:

- **Insufficient training data:** Only 87 training samples (1.3% of training set). With 5 classes and 12 stellar features, the model lacks the inductive bias to learn reliable decision boundaries from so few examples.
- **Light-curve ambiguity:** An ephemeris match can produce a transit-shaped dip that is morphologically indistinguishable from a genuine planet transit. The discriminating information is the period's coincidence with a catalog entry, which is an indirect signal encoded in koi\_period — not directly observable from the light curve shape.

We attempted to improve Ephemeris Match performance by increasing the sampler boost to 8x and adding a selective loss weight of 5x for class 4 only. While Ephemeris F1 improved marginally from 0.071 to 0.113, the intervention caused Stellar Eclipse recall to drop from 66.5% to 54.7% — a net negative trade-off given that Stellar Eclipse has 17x more test samples. The experiment demonstrates that loss-engineering cannot substitute for adequate labeled data in the extreme imbalance regime.

**Upper bound estimate:** Given 87 training samples and 5-class decision boundaries, achieving  $F1 > 0.3$  for Ephemeris Match would likely require a minimum of 500–1000 additional labeled examples, or the incorporation of catalog cross-matching features (e.g., whether the period is within 1% of any known KOI) as a hard-coded feature to the stellar branch.

## C. Lessons Learned

- **One imbalance correction per axis:** Data-level (Sampler) and loss-level (Focal/weighted CE) corrections should not be stacked — they operate on the same underlying signal and compound non-linearly.
- **Focal Loss > Weighted CE for multi-class imbalance:** The hard-example focusing mechanism of Focal Loss is more principled than inverse-frequency weighting when class overlap is high, as it directly modulates gradient contribution by prediction difficulty.
- **Multi-task learning provides regularisation:** The regression and confidence heads prevent the classification head from collapsing to trivial solutions by enforcing physical consistency constraints on transit parameters.
- **Cache preprocessing:** Light curve preprocessing (phase folding + binning) is deterministic and expensive. Caching reduces per-epoch time by  $\sim 12\times$  (166s  $\rightarrow$  14s) with no loss of training fidelity.

## VIII. CONCLUSION AND FUTURE WORK

We presented AstroLight, a multi-task late-fusion deep learning system for five-class exoplanet transit signal classification on the NASA Kepler KOI catalog. Our best model achieves 79.93% five-class accuracy and 0.8041 weighted F1, with a transit F1 of 0.9254 and a substantially improved Centroid Offset F1 of 0.6238 compared to the CrossEntropy baseline. We provided a detailed analysis of two critical failure modes: the centroid offset gravity well (resolved via Focal Loss) and the Ephemeris Match data starvation problem (unresolved — a fundamental data limitation).

The project trajectory — from hackathon binary classifier to multi-task multi-class research system — underscores that problem statement evolution is as important as architectural innovation. Each expansion of the classification granularity revealed new failure modes that drove corresponding improvements in training strategy.

### Future Work

- **Ephemeris Match augmentation:** Generate synthetic Ephemeris Match samples by re-labeling KOIs whose periods are within 1% of known planets — potentially tripling the available training data.
- **Transformer-based light curve encoder:** Replace 1D-ResNets with a phase-aware transformer that can model long-range dependencies in the phase-folded sequence.
- **Graph neural network for catalog cross-referencing:** Encode the Kepler catalog as a period-similarity graph; Ephemeris Match nodes would be naturally distinguishable by their high-degree connectivity.
- **TESS generalisation:** Evaluate AstroLight on TESS TCEs to assess cross-mission transfer, given TESS's shorter baselines and different noise profile.
- **Ablation study:** Systematically remove each branch (Global only, Local only, Stellar only, Global+Local) to quantify each modality's contribution.

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